

Proof of Smoothness for Morphic-Regularized Navier-Stokes Equations

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September 6, 2024

Step 1: Ensuring Non-Singularity

The first step toward proving smoothness is to ensure that the introduction of the morphic function prevents the formation of singularities. Singularities in fluid dynamics, such as those occurring in a vortex, arise when the velocity field develops infinite gradients, leading to unbounded behavior. By introducing a morphic regularization, we aim to modify the vortex core and other regions where singularities might form, smoothing out the velocity field and keeping the system well-behaved.

Smoothing with the Morphic Function

The morphic function introduces a regularization parameter ϵ that acts as a small buffer or smoothing term. This regularization prevents the velocity field from becoming infinite at critical points, such as the center of a vortex.

Consider the classical case of a vortex where the velocity field is given by:

$$u(r) = \frac{1}{r},$$

where r is the radial distance from the vortex center. As $r \rightarrow 0$, the velocity $u(r)$ diverges to infinity, creating a singularity at the vortex core.

To prevent this singularity, we modify the velocity field using the morphic regularization term ϵ , giving:

$$u_{\text{morphic}}(r) = \frac{1}{\sqrt{r^2 + \epsilon^2}},$$

where ϵ is a small positive constant that ensures the velocity remains finite as $r \rightarrow 0$.

Explanation of the Regularization Term

The regularization term ϵ^2 acts as a buffer, preventing the velocity from becoming infinite at $r = 0$. Specifically:

$$u_{\text{morphic}}(0) = \frac{1}{\epsilon},$$

which remains finite as long as ϵ is non-zero. The presence of ϵ smooths out the velocity field near the singularity, ensuring that the velocity does not diverge.

For larger values of r , the regularized velocity field $u_{\text{morphic}}(r)$ behaves similarly to the classical vortex solution. As r increases, the influence of the ϵ^2 term diminishes, and the velocity field asymptotically approaches:

$$u_{\text{morphic}}(r) \approx \frac{1}{r} \quad \text{for } r \gg \epsilon.$$

Thus, the morphic regularization only significantly affects the region close to $r = 0$, where it prevents singular behavior.

Implications for Fluid Dynamics

The morphic regularization provides a systematic way to smooth out singularities in the velocity field, particularly in regions like vortex cores where classical solutions predict unbounded behavior. By introducing the regularization term, we ensure that:

- The velocity field remains bounded for all values of r ,
- Singularities at $r = 0$ are eliminated,
- The modified velocity field behaves similarly to the classical solution at larger distances, preserving the key characteristics of the vortex outside the singular region.

This regularization technique can be generalized to other fluid systems prone to singularities, providing a framework for maintaining smooth solutions in previously problematic cases.

Conclusion of Step 1

By introducing the morphic function and the regularization parameter ϵ , we ensure that the velocity field remains finite, preventing the formation of singularities. This step lays the foundation for the overall proof of smoothness, as it addresses one of the key challenges in fluid dynamics: controlling the behavior of the system near singular points.